

Project Report

Active-Loaded Common Source Amplifier (Cadence Virtuoso)

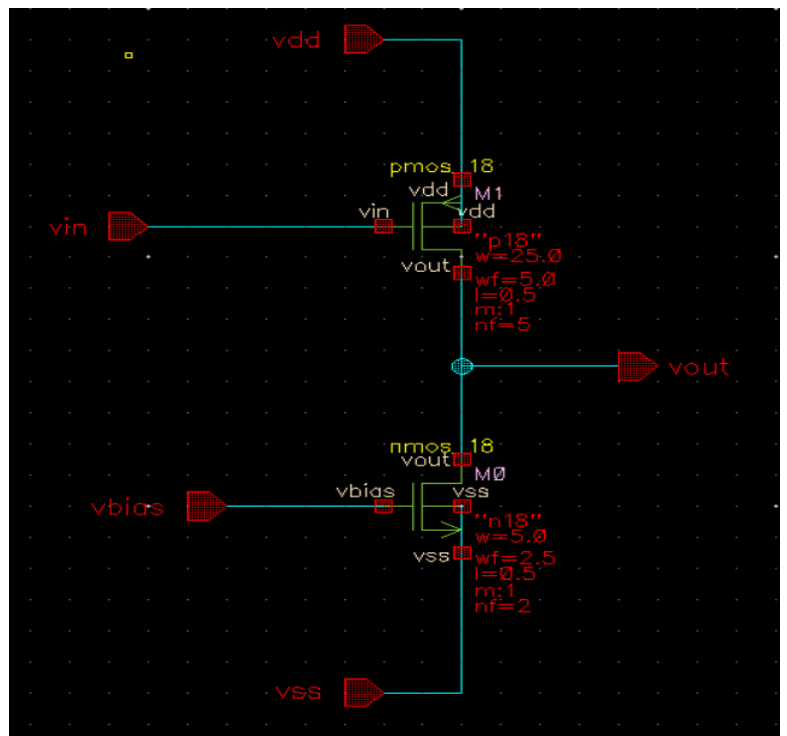
Project Overview: Designed an SCL 180nm analog layout achieving 36 dB gain through multi-finger transistors, with clean DRC/LVS and Parasitic Extraction (PEX) using Siemens Calibre.

1. Introduction

The objective of this project was to design, simulate, and lay out an active-loaded Common Source (CS) amplifier using the SCL 180nm CMOS technology. A PMOS active load was utilized to provide a high output impedance, thereby maximizing the voltage gain of the amplifier. To optimize layout area and reduce parasitic gate resistance and capacitance, a multi-finger transistor geometry was implemented.

2. Schematic Design

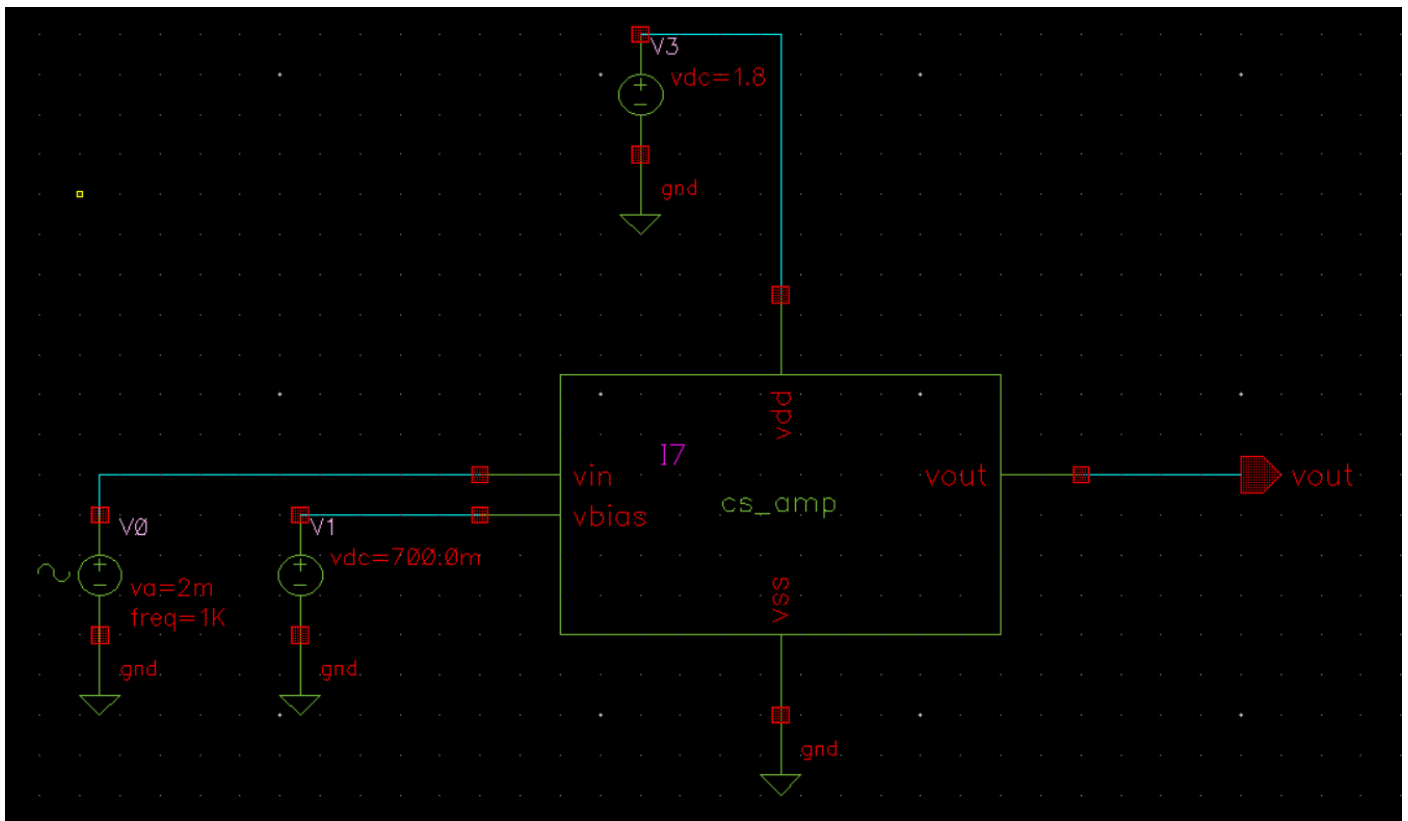
The core amplifier consists of an NMOS driving transistor and a PMOS load. The parameters were carefully tuned to meet the target specifications. The PMOS active load uses 5 fingers, each with dimensions of $W = 5.0 \mu\text{m}$ and $L = 0.5 \mu\text{m}$. The NMOS input transistor utilizes 2 fingers, each with dimensions of $W = 2.5 \mu\text{m}$ and $L = 0.5 \mu\text{m}$. A testbench was constructed with a 1.8V VDD and a DC bias of 700mV applied to the NMOS gate.



Schematic of a CS amp with parameters

3. Simulation Results

Pre-layout were performed to verify the DC operating point, frequency response (AC), and time-domain behavior (Transient) for DC gain of the Amplifier



Testbench setup with VDD and bias voltage.

3.1 Transient Response

Applying a 1kHz sinusoidal input signal shows a clean, amplified, and 180-degree phase shifted output waveform, which is the hallmark of an inverting common-source topology.

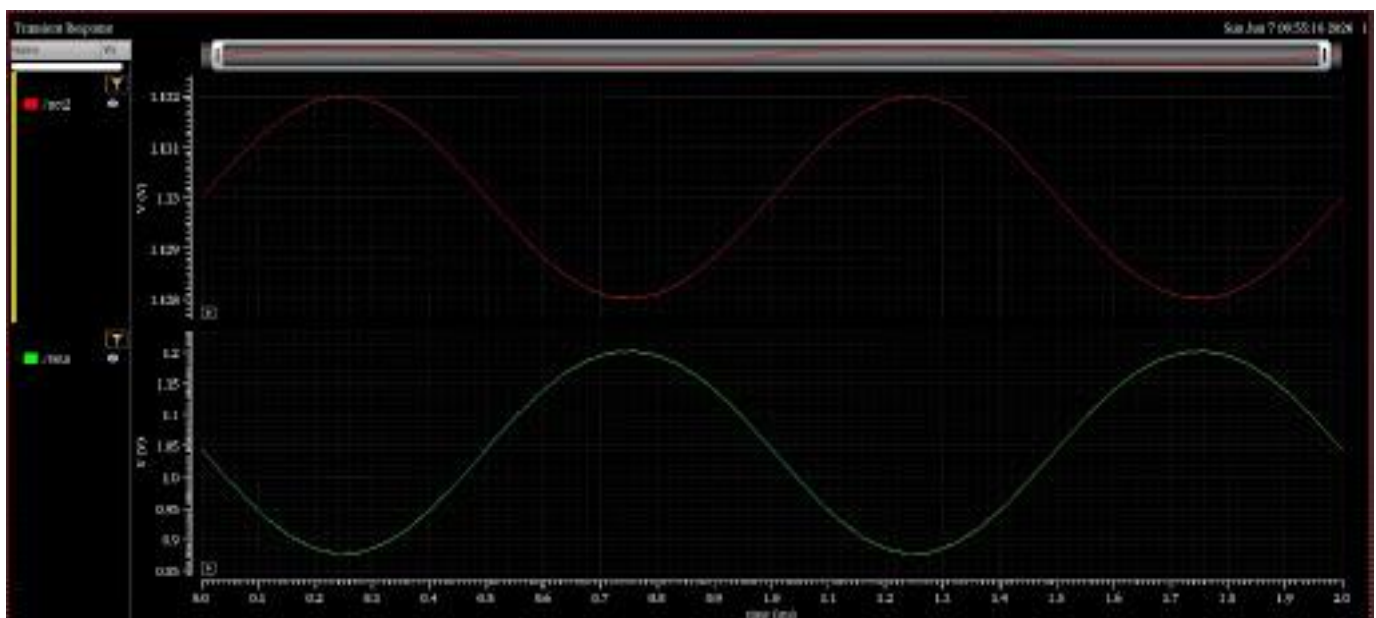
Calculation of Voltage Gain

Input Signal ($V_{in,pp}$): $1.1320 \text{ V} - 1.1280 \text{ V} = 4 \text{ mV}_{pp}$

Output Signal ($V_{out,pp}$): $1.200 \text{ V} - 0.875 \text{ V} = 325 \text{ mV}_{pp}$

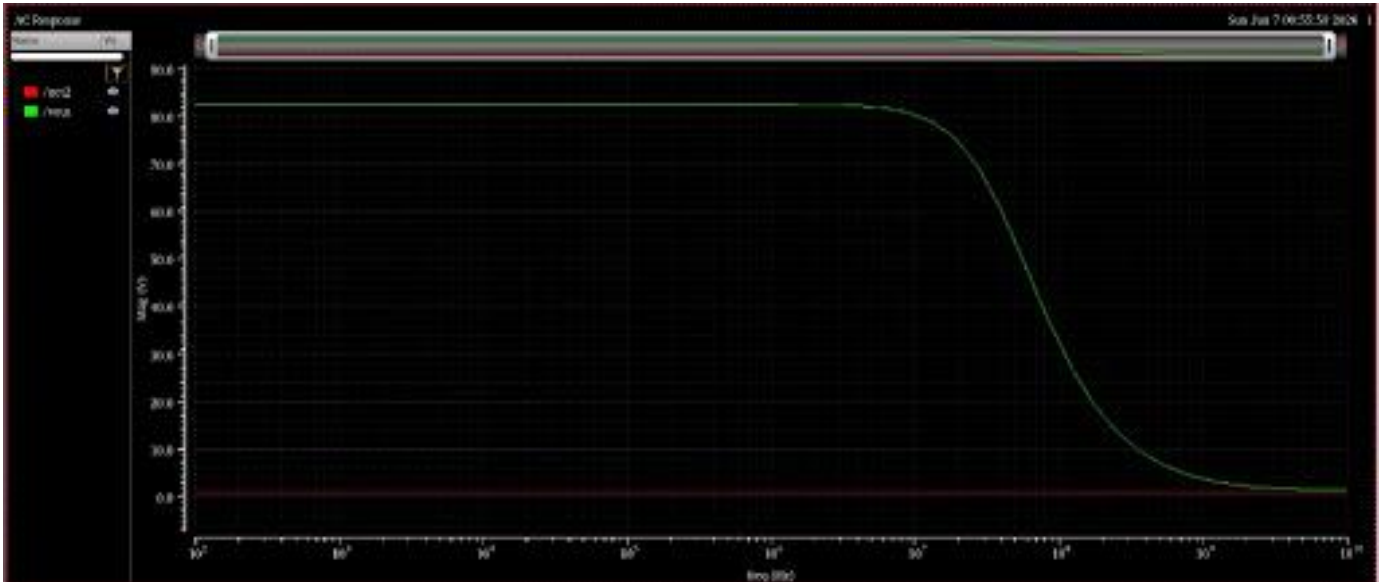
$A_v = V_{out,pp} / V_{in,pp} \Rightarrow$ This is the voltage gain formula

$A_v = 325 \text{ mV} / 4 \text{ mV} = 81.25 \text{ V/V} \Rightarrow A_v(\text{dB}) = 20 * \log_{10}(81.25) \Rightarrow \sim 36 \text{ dB}$



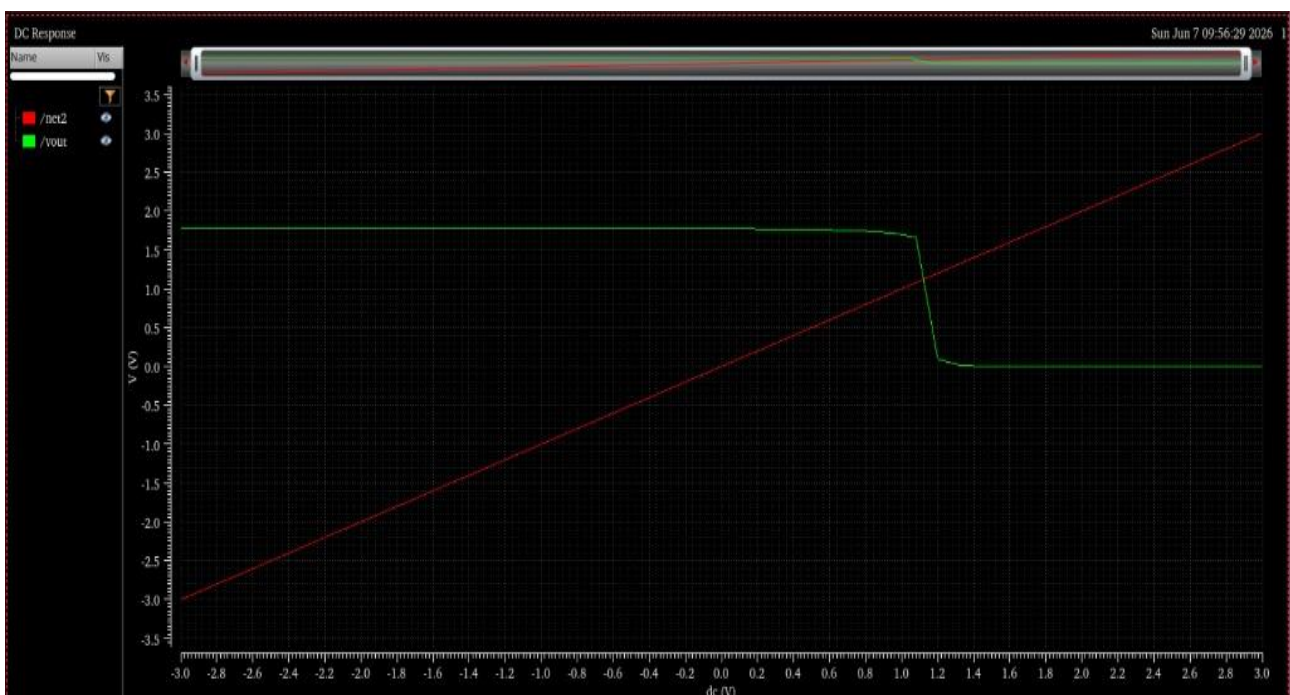
3.2 AC Response

The frequency response plot demonstrates that the amplifier achieves a peak DC gain of approximately 36 dB , ~ 82 V/V . The gain remains flat across the low-frequency band before rolling off at higher frequencies due to internal parasitic capacitances



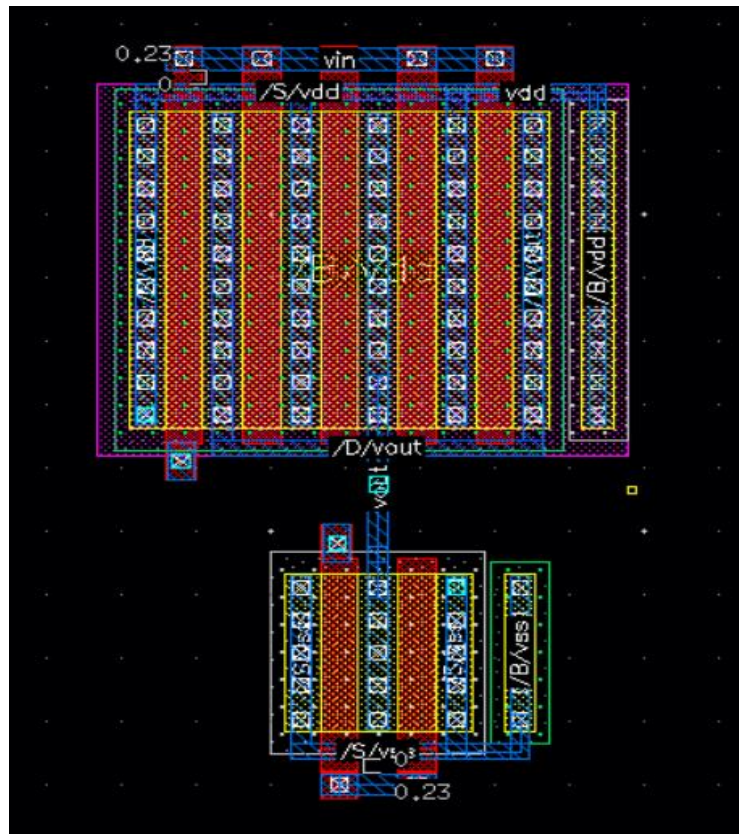
3.3 DC Response

The DC sweep analysis confirms the transistors are correctly biased in the saturation region, ensuring the amplifier operates within its high-gain linear regime. The DC operating point for the input node is set at 1.13 V, which provides the necessary gate-to-source voltage to maintain the NMOS in saturation. This optimal bias point ensures the amplifier maintains maximum linear signal swing and stable performance for the target 36 dB gain.



4. Physical Layout and Verification

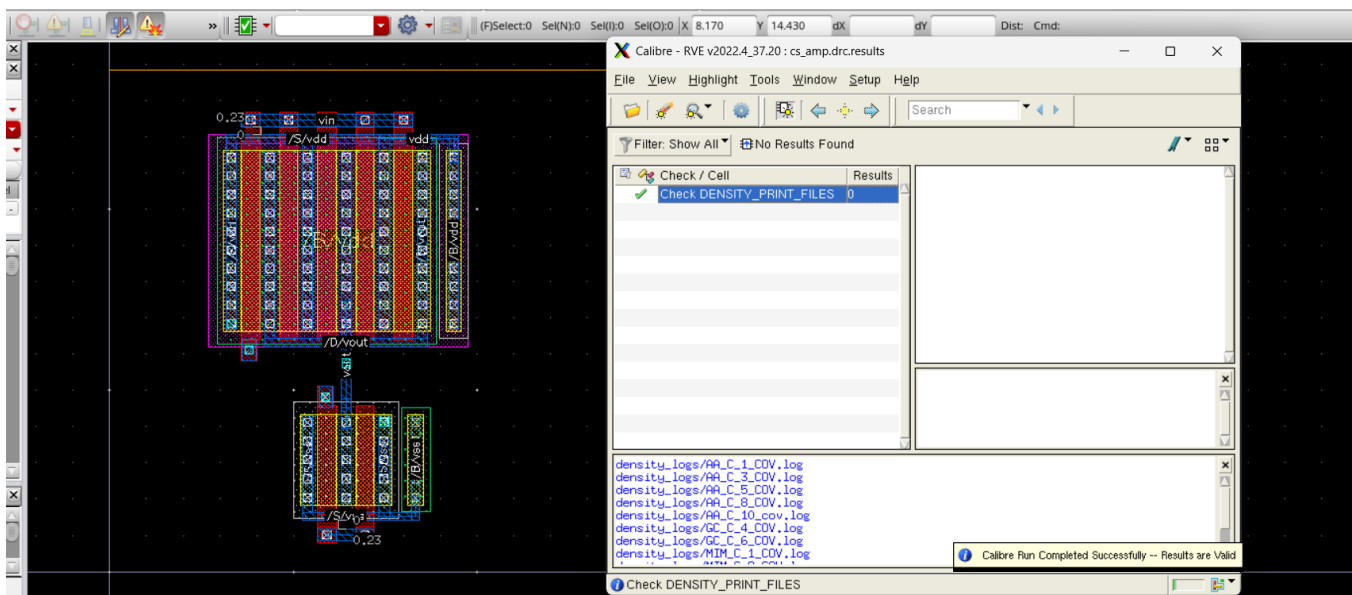
The layout was designed in Cadence Virtuoso using the SCL 180nm PDK, employing multi-finger transistors to optimize area and minimize parasitic gate resistance. Following the layout, verification was performed using Siemens Calibre. The design successfully passed the Design Rule Check (DRC) with zero violations, and the Layout vs. Schematic (LVS) verification confirmed a perfect match between the drawn layout and the schematic. Finally, Parasitic Extraction (PEX) was executed to generate a netlist including parasitic RC components for accurate post-layout simulation.



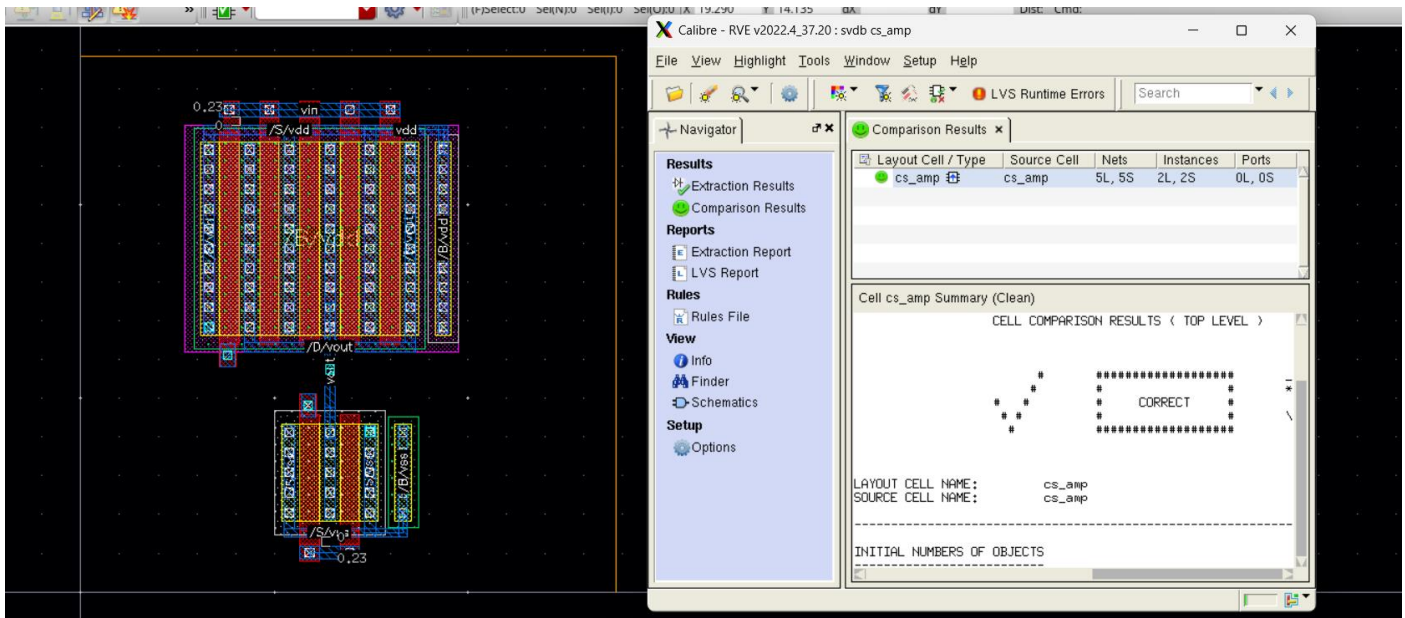
Layout of a CS_amp, Top (PMOS) bottom (NMOS)

5. Design Rule Check (DRC) & Layout vs. Schematic (LVS)

The layout was verified using Siemens Calibre. It successfully passed the Design Rule Check (DRC) with zero violations, ensuring it meets all foundry manufacturing constraints. Following DRC, Layout vs. Schematic (LVS) verification was executed, confirming a perfect topological and parameter match between the drawn schematic and the layout. Finally, Parasitic Extraction (PEX) was run to generate the netlist containing parasitic RC components for accurate post-layout simulations.



DRC clean , no errors in the right side calibre - RVE panel

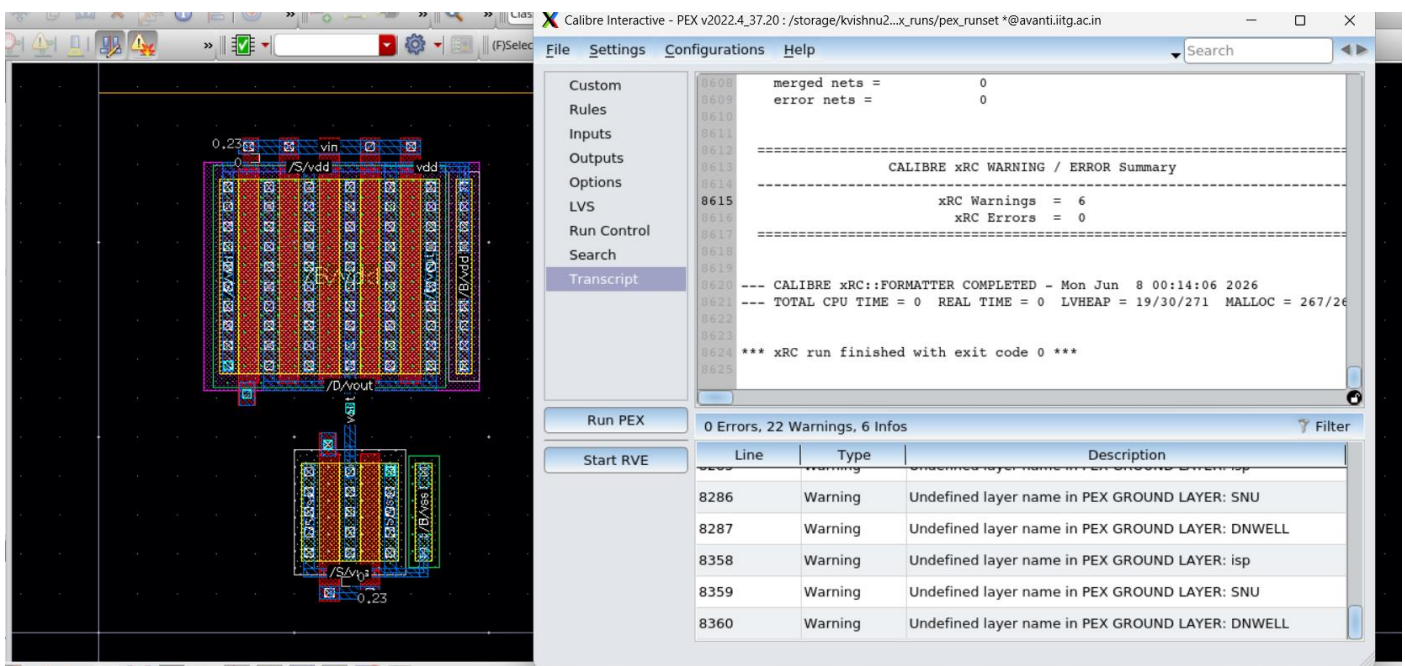


LVS clean , big smiley face in the right panel says it

Statistics:

- 1 layout instance was filtered and its pins removed from adjoining nets.
- 7 layout mos transistors were reduced to 2 , 5 mos transistors were deleted by parallel reduction.

This says that our finger division of Transistors worked out perfectly and the fingers were merged into respective Transistors



PEX from calibre is also done

After verifying the layout connectivity via LVS, Parasitic Extraction (PEX) was performed using Siemens Calibre. This process identifies the hidden parasitic resistances R and capacitances C caused by the physical routing of metal wires and vias , after which Post layout simulations will be performed for exact results

Sample .pex.netlist file for RC parasitic values

```
mr_ni "VOUT" 253.503 0 3.41043e-15 '( "MM0@2_d" "MM0_d" "MM1@2_d"  
"MM1@3_d" "MM1@4_d" "MM1@5_d" "MM1_d" )
```

- `mr_ni "VOUT"`: Defines this as the "Net Information" (summary) for the node named VOUT.
- `253.503`: This is the total resistance (in Ohms) of the entire VOUT net.
- `0`: This is the total conductance
- `3.41043e-15`: This is the total parasitic capacitance (in Farads) of the net—approximately 3.41 femtofarads.
- `'( ... )`: This list contains all the "driving points" (transistor drains) connected to this node.

```
mr_pp 'r "rVOUT_48" ("VOUT_82" "VOUT_84") 14.7292
```

This indicates a parasitic resistor named `rVOUT_48` exists between internal nodes `VOUT_82` and `VOUT_84` with a resistance of 14.73 ohm.

```
mr_pp 'r "rVOUT_49" ("VOUT_82" "VOUT") 18.6667
```

This indicates a resistor named `rVOUT_49` exists between internal node `VOUT_82` and your final output terminal `VOUT` with a resistance of 18.67 ohm.

6. Conclusion

An active-loaded Common Source amplifier was successfully designed, simulated, and physically laid out using the SCL 180nm CMOS process. Pre-layout simulations confirmed that the circuit met its target specifications, achieving a voltage gain of approximately 36 dB (81.25 V/V) while maintaining a stable DC operating point at 1.13 V to ensure optimal saturation and maximum linear signal swing.

To optimize silicon area and minimize parasitic gate resistance, a multi-finger transistor topology was effectively implemented. The structural integrity and manufacturability of the physical layout were rigorously validated using Siemens Calibre. The design achieved a clean DRC with zero violations and a flawless LVS match, which successfully verified the parallel finger reductions. Finally, Post-Layout Parasitic Extraction (PEX) confirmed a highly efficient metal routing strategy, yielding a total output path parasitic resistance of only 253.5 Ohms and a minimal capacitance of 3.41 fF.